

Particle filter concepts



CHALMERS
UNIVERSITY OF TECHNOLOGY

Particle filters are useful as,

Green: they can handle almost any models.

Orange: within reasonable computational complexity, the particle filter always gives us the best performance.

Pink: they give us a compact description of the posterior density.

Yellow: give as a non-parametric description of the posterior.

With the Particle filters we approximate the posterior as

$$p(\mathbf{x}_k | \mathbf{y}_{1:k}) \approx \sum_{i=1}^N w_k^{(i)} \delta(\mathbf{x}_k - \mathbf{x}_k^{(i)})$$

Which of the following statements are true regarding this approximation?

Green: $\mathbf{x}_k$ can take any value in $[\min_i(\mathbf{x}_k^{(i)}), \max_i(\mathbf{x}_k^{(i)})]$.

Orange: We can view this as a discrete distribution, where $P\{\mathbf{x}_k = \mathbf{x}_k^{(i)} | \mathbf{y}_{1:k}\} = w_k^{(i)}$.

Pink: It follows that, $\mathbb{E}\{\mathbf{x}_k | \mathbf{y}_{1:k}\} \approx \sum_{i=1}^N w_k^{(i)} \mathbf{x}_k^{(i)}$.

Yellow: We can get arbitrary fine approximation by just increasing the number of particles, N .